Practice Exam - 1

Question 77 of 117

A 36-year-old woman is two years status post-transplant for a titin-related cardiomyopathy. Preoperatively, she was a blood type O and had a calculated PRA of 54% using an MFI threshold of 5000. Following transplant at 4 weeks, she developed two donor-specific donor antibodies (B57 at an MFI of 3065 and DR52 at an MFI of 2518) without signs of graft dysfunction by echo or hemodynamics and biopsies have shown ISHLT CMR grade 0-1R and AMR 0. By 6 months post-transplant, both donor specific antibodies disappeared. Her last echo a year ago showed an LVEF of >65% and no valve dysfunction. Her drug regimen includes tacrolimus (level 6-8) and myfortic 720 mg twice a day. She has been a compliant patient.

The patient was admitted to your hospital today because of severe shortness of breath and feeling like her heart is “racing”. Blood pressure 90/70, HR 130 bpm. An echocardiogram was performed showing absence of LV dilation but an ejection fraction of 35-40% and thick walls (IVS 18 mm). She underwent urgent endomyocardial biopsy and the catheterization lab called you with the following hemodynamics: RA 14, PA 50/30, WP 30, cardiac index 1.6. Cr 1.5 (up from 1.3), hsTroponin 390, NT-BNP 5275, lactate 1.5.

The next appropriate step while awaiting pathology results would be to:

A. Send donor specific antibodies and begin IVIG and pheresis

B. Send donor specific antibodies and treat with 1g IV solumedrol and anti-thymocyte globulin [rabbit] (ie. thymoglobulin)

C. Send donor specific antibodies and treat with an IL-2 R antagonist (e.g. basiliximab)

D. Send donor specific antibodies and begin evaluation for redo heart transplant

Commentary | References

Content Code (Blueprint) - Select the appropriate domain and subdomain from the AHFTC blueprint:

Task (Choose one category: Diagnosis / Management / Test Selection / Interpretation / Prognosis): Management

Teaching Point: This patient presented with clinical symptoms concerning for graft failure. While a diagnosis of AMR and/or CMR was not clearly provided in the vignette, treatment cannot be delayed while awaiting results of pathology.

Correct Answer (Best Answer Choice): B.

Distractors (3–4 Plausible Incorrect Options):

1. While this patient certainly could have AMR, the first line of treatment for severe AMR and/or CMR is steroids and cytolytics. Pheresis and IVIG and/or other immune modulating drugs (e.g. rituximab, Eculizumab, etc) can be added on after biopsy results and donor specific antibody results are availaby, especially if there is lack of response to steroids/cytolytics.
2. Correct
3. While donor specific antibody (DSAb) assessment is needed, there is no role for Il2-R antagonist as a first line treatment in this patient. First line therapy is steroids and cytolytics, which are amended based on biopsy results and results of donor specific antibody
4. While donor specific antibody (DSAb) assessment is needed, initiation of a transplant evaluation without attempting to treat rejection is potentially deadly. Patient needs to be stabilized, pre-emptive therapy started to address rejection that may be present, all of which is followed by treatment directed at final diagnosis achieved from biopsy and additional subsequent testing.

Rationale – This patient presented with clinical symptoms and a moderate LVEF decline. Hemodynamics from right heart catheterization revealed the true severity of graft compromise. While a diagnosis of AMR and/or CMR was not clearly provided in the vignette, treatment cannot be delayed while awaiting results of pathology. For all answers, sending donor specific antibodies is correct but these results also take time to return. This patient should receive high dose IV steroids regardless of ISHLT endomyocardial biopsy grade (class I recommendation) and cytolytics are a reasonable adjunct therapy. Treatment is then tailored to results of biopsies (CMR and AMR findings),donor specific antibodies, and response to initial steroid/cytolytic therapy.